



KAMBALA

Student Number: _____

Task 4
September 2014

Preliminary HSC Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours and 30 minutes
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- Show all necessary working in Questions 9 - 13

Total marks – 83

Section I
8 marks

- Attempt Questions 1 - 8
- Allow about 15 minutes for this section

Section II
75 marks

- Attempt Questions 9 - 13
- Allow about 2 hours and 15 minutes for this section

Section I

8 Marks

Attempt Questions 1 – 8

Allow about 15 minutes for this section

Use the answer sheet for Questions 1 – 8.

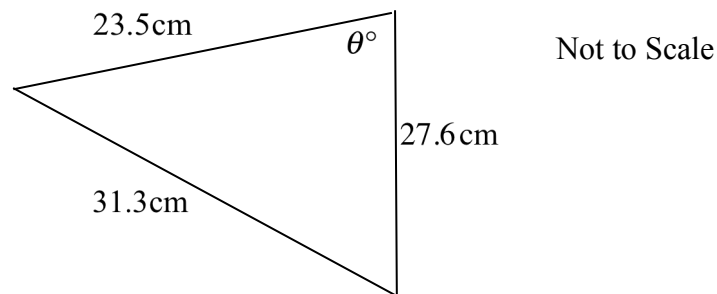
1. What is $\frac{2}{3-\sqrt{2}}$ as a fraction with a rational denominator?
 - (A) $\frac{6-2\sqrt{2}}{7}$
 - (B) $\frac{6+2\sqrt{2}}{7}$
 - (C) $6-2\sqrt{2}$
 - (D) $6+2\sqrt{2}$

2. What is the value of $f(-1)$ if $f(x) = x^2 - 4x$?
 - (A) $f(-1) = -3$
 - (B) $f(-1) = -5$
 - (C) $f(-1) = 3$
 - (D) $f(-1) = 5$

3. Select the axis of symmetry of the parabola $y = 5x^2 - 4x - 3$.
 - (A) $x = \frac{4}{5}$
 - (B) $x = \frac{2}{5}$
 - (C) $x = -\frac{3}{5}$
 - (D) $x = -\frac{4}{5}$

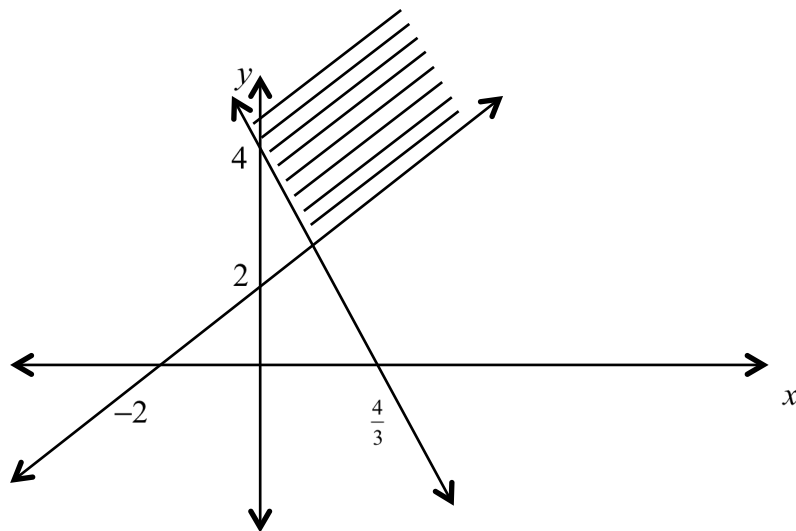
4. Which of the following is true for the equation $3x^2 - 5x - 2 = 0$?
- (A) no real roots
- (B) one real root
- (C) two rational distinct roots
- (D) two irrational distinct roots.
5. Find the value of b if $\sin 2b = \cos (b + 30^\circ)$
- (A) 20°
- (B) 30°
- (C) 40°
- (D) 50°
6. Which of the following is **NOT** a correct expression involving θ in $\triangle ABC$?

1



- (A) $31.3^2 = 27.6^2 + 23.5^2 - 2 \times 27.6 \times 23.5 \cos \theta$
- (B) $\cos \theta = \frac{23.5^2 + 27.6^2 - 31.3^2}{2 \times 23.5 \times 27.6}$
- (C) $\frac{31.3}{\sin \theta} = \frac{27.6}{\sin 58^\circ 26'}$
- (D) $\frac{\sin \theta}{31.3} = \frac{\sin 58^\circ 26'}{23.5}$

7. Which pair of inequalities describes the shaded region?



- (A) $y \geq 4 - 3x$
 $y \geq x + 2$
- (B) $y \geq 4 - 3x$
 $y \leq x + 2$
- (C) $y \leq 4 - 3x$
 $y \geq x + 2$
- (D) $y \leq 4 - 3x$
 $y \leq x + 2$
8. If $\operatorname{cosec} \theta = -\frac{5}{3}$ and $\cos \theta > 0$, find $\cot \theta$.

- (A) $\cot \theta = \frac{3}{4}$
- (B) $\cot \theta = \frac{4}{3}$
- (C) $\cot \theta = -\frac{3}{4}$
- (D) $\cot \theta = -\frac{4}{3}$

End of Section I

Section II

75 Marks

Attempt Questions 9 - 13

Allow about 2 hours 15 minutes for this section

Answer each question on the writing paper provided. Start each question on a new page.

In Questions 9 - 13, your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (15 marks)

- (a) Evaluate $\sqrt{\frac{2.3^2 + 3.7^2}{7.5}}$ correct to 2 significant figures. 2
- (b) Factorise $x^3 - 125$. 1
- (c) Solve $|2x - 3| = 7$ for x . 2
- (d) Consider the function $f(x) = x^2 + 2x - 3$.
- (i) Find the x and y intercepts. 2
- (ii) Find the minimum value of the function. 2
- (iii) Sketch the function. 1
- (iv) For what values of x is the curve decreasing? 1
- (e) Sketch the locus of the point that moves so that it is always 2 units from the point $P(4, 2)$. 2
- (f) If $f(x) = x^2 + 3x - 10$ find:
- (i) $f'(x)$. 1
- (ii) the value of x for which $f'(x) = 0$. 1

Question 10 (15 marks) Start a new page.

(a) Find the exact value of $\tan 300^\circ$. 2

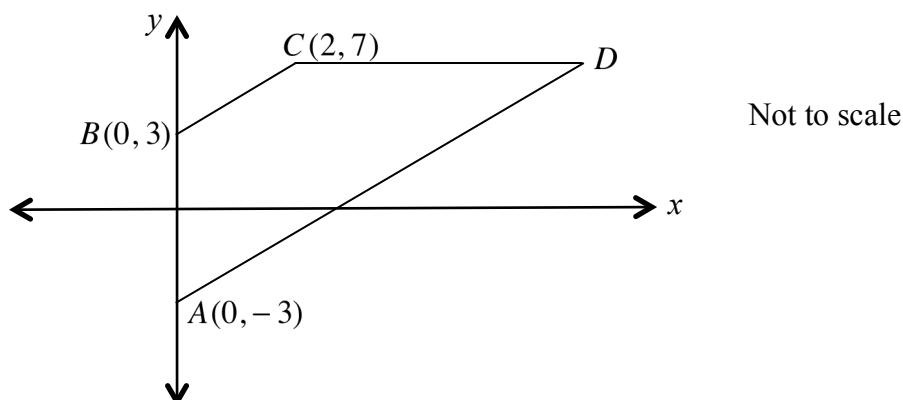
(b) In solving a quadratic equation, a student wrote down $x = \frac{4 \pm \sqrt{16 + 96}}{6}$. 2

What was the original quadratic equation?

(c) (i) Solve the inequality $(x + 10)(x + 2) \geq 0$ for x . 2

(ii) Hence state the domain of $\sqrt{x^2 + 12x + 20}$. 1

(d) In the diagram, $ABCD$ is a quadrilateral. The equation of the line AD is $2x - y - 3 = 0$.



(i) The line CD is parallel to the x -axis. Find the coordinates of D . 1

(ii) Show that $ABCD$ is a trapezium by showing that BC is parallel to AD . 2

(iii) Find the exact lengths of BC and AD . 2

(iv) Show that the perpendicular distance from B to AD is $\frac{6}{\sqrt{5}}$ units. 2

(v) Hence find the area of the trapezium $ABCD$. 1

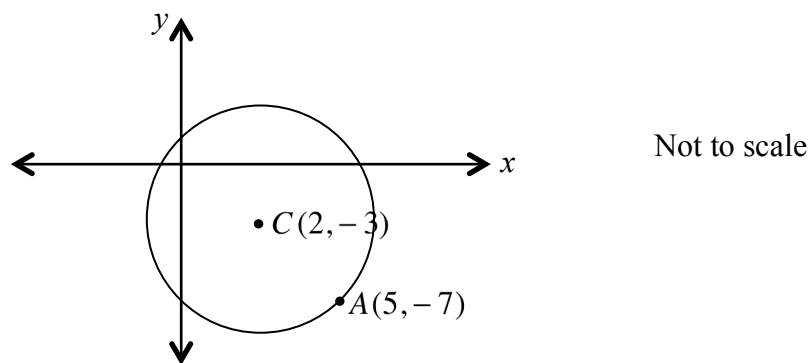
Note: $A = \frac{1}{2}(a + b)h$

Question 11 (15 marks) **Start a new page.**

(a) Simplify $\frac{m^2 - 4}{mn} \times \frac{2m}{2m - 4}$. 2

(b) Explain why $x^2 + 6x + 11$ is positive definite. 2

(c) A circle with centre $C(2, -3)$ has one end of a diameter at $A(5, -7)$. Find the coordinates of the other end of the diameter. 2



(d) Let α and β be the roots of the equation $3x^2 + 5x - 1 = 0$. Find:

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $(\alpha + 1)(\beta + 1)$ 2

(e) Differentiate the following with respect to x :

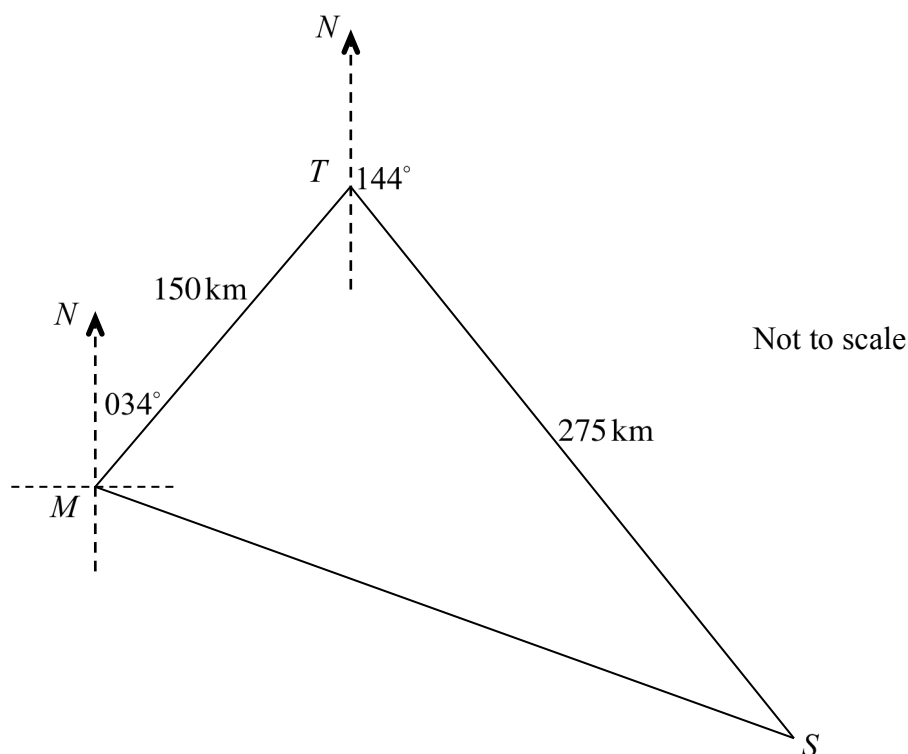
(i) x^{-3} 1

(ii) $(x - 3)(2x + 1)$ 2

(iii) $\frac{5x^6 - 7x^2}{x}$ 2

Question 12 (15 marks) Start a new page.

- (a) A parabola has focus $S(2, -1)$ and the equation of the directrix is $x = -4$.
- (i) Mark this information on a diagram and find the coordinates of the vertex. 2
- (ii) Write down the equation of the parabola. 1
- (b) Solve the equation $(x^2 - 2)^2 - 4(x^2 - 2) - 21 = 0$ for x . 3
- (c) A ship sails from Melbourne, M , for 150 kilometres on a bearing of 034° to point T . It then sails on a bearing of 144° for 275 kilometres to point S as shown in the diagram below.



Copy the above diagram neatly onto your page.

- (i) Show that $\angle MTS = 70^\circ$. 1
- (ii) How far, to the nearest kilometre, is the ship at point S , from Melbourne, M ? 2
- (iii) What is the bearing of the ship, S , from Melbourne, M , to the nearest degree? 2

Question 12 continued

(d) A function is given by $f(x) = x^2 - 1$.

(i) Find $f(2)$ and $f(2+h)$. 2

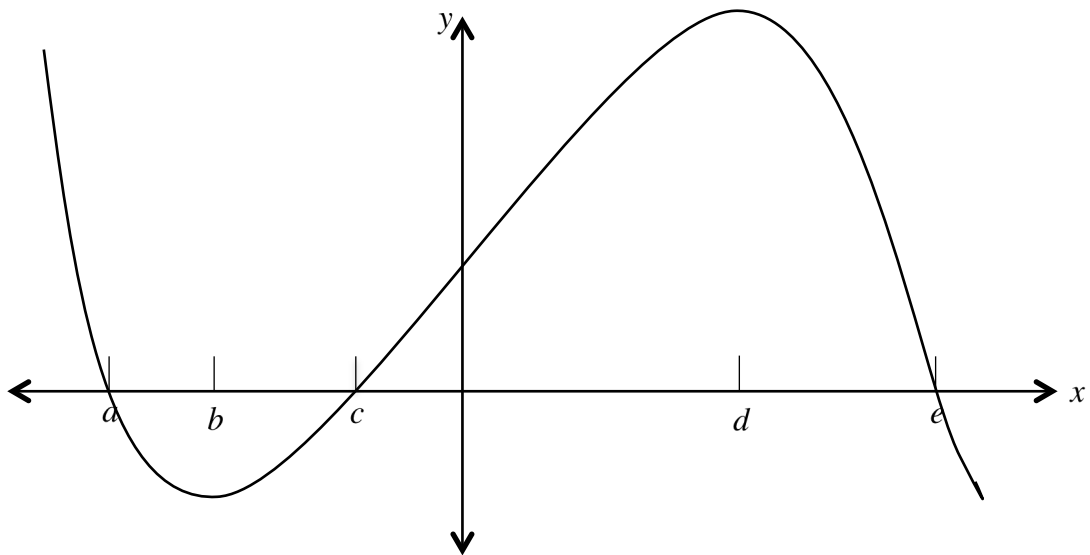
(ii) Show that $\frac{f(2+h) - f(2)}{h} = 4 + h$. 1

(iii) Hence, or otherwise, find $f'(2)$. 1

Question 13 (15 marks) Start a new page.

(a) If $2x^2 - 3x - 4 \equiv A(x+2)^2 + B(x+2) + C$ for all values of x , find the values of A , B and C . 3

(b) The curve below represents the graph of function $y = f(x)$.



(i) Is the function odd, even or neither? 1

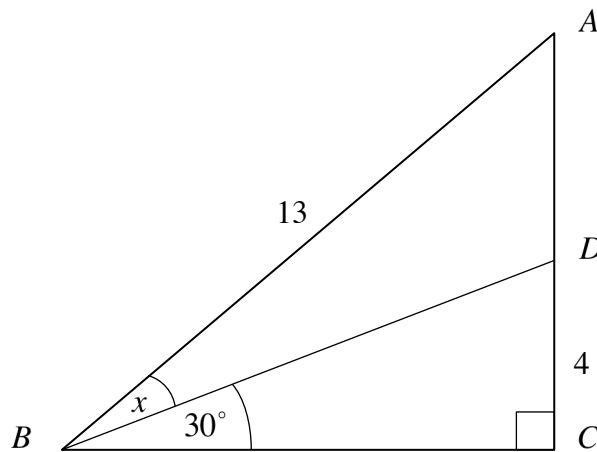
(ii) At what point(s) is $f(x) = 0$? 1

(iii) Where is the gradient of the function equal to zero? 1

(iv) Where is the gradient of the function positive? 1

Question 13 continued

- (c) The right-angled triangle ABC has hypotenuse $AB = 13$. The point D is on AC such that $DC = 4$, $\angle DBC = 30^\circ$ and $\angle ABD = x$.



Not to scale

- | | | |
|------|---|----------|
| (i) | Find the length of BD . | 1 |
| (ii) | Using the sine rule, or otherwise, find the exact value of $\sin x$. | 3 |
-
- | | | |
|------|---|----------|
| (d) | (i) If the line $y = x + m$ cuts the circle $x^2 + y^2 = 4$, show that the x -coordinates of the points of intersection can be found by solving $2x^2 + 2mx + m^2 - 4 = 0$. | 1 |
| (ii) | For what value(s) of m will the line $y = x + m$ be a tangent to the circle? | 3 |

End of Section II

Kambala Preliminary HSC Mathematics

Assessment Task 4

September 2014

SOLUTIONS

$$1. \frac{2}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}}$$

$$= \frac{6+2\sqrt{2}}{7} \quad \textcircled{B}$$

$$2. f(x) = x^2 - 4x$$

$$f(-1) = (-1)^2 - 4(-1)$$

$$= 1 + 4$$

$$= 5 \quad \textcircled{D}$$

$$3. y = 5x^2 - 4x - 3$$

$$x = \frac{-b}{2a}$$

$$x = \frac{4}{2(5)}$$

$$x = \frac{2}{5} \quad \textcircled{B}$$

$$4. 3x^2 - 5x - 2 = 0$$

$$\Delta = (-5)^2 - 4(3)(-2)$$

$$= 25 + 24$$

$$= 49$$

$\therefore \Delta > 0 \Rightarrow$ two distinct roots

Δ is a perfect square

$\Rightarrow$ rational roots $\textcircled{C}$

$$5. \sin 2b = \cos(b+30)$$

$$2b + b + 30 = 90^\circ$$

$$3b + 30 = 90$$

$$3b = 60$$

$$b = 20^\circ \quad \textcircled{A}$$

$$6. D \quad \text{(Using Cosine Rule, angle b/w 23.5 and 31.3 is } 58^\circ 26')$$

$$7. y > 4 - 3x$$

test (0,0)

$$0 > 4$$

false

$\therefore$ region above $y = 4 - 3x$

$$\therefore y > 4 - 3x$$

$$y > x + 2$$

test (2,5)

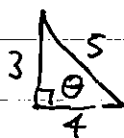
$$5 > 2 + 2$$

true

$$\therefore y > 4 - 3x, y > x + 2 \quad \textcircled{A}$$

$$8. \operatorname{cosec} \theta = -\frac{5}{3}, \cos \theta > 0$$

Quadrant 4



$$\cot \theta = -\frac{4}{3} \quad \textcircled{D}$$

Question 9

$$a) \sqrt{\frac{2 \cdot 3^2 + 3 \cdot 7^2}{7.5}}$$

$$= \sqrt{\frac{5 \cdot 29 + 13 \cdot 69}{7.5}}$$

$$= \sqrt{\frac{18 \cdot 98}{7.5}}$$

$$= \sqrt{2.5306}$$

$$\approx 1.5908$$

$$= 1.6 \text{ (to 2 sig figs)}$$

$$b) x^3 - 125$$

$$= (x-5)(x^2+5x+25)$$

$$c) |2x-3|=7$$

$$2x-3=7 \text{ or } 3-2x=7$$

$$2x=10$$

$$x=5$$

$$-2x=4$$

$$x=-2$$

$$\therefore x = -2, 5$$

$$d) f(x) = x^2 + 2x - 3$$

$$i) \text{ when } x=0, y=-3$$

$$\text{when } f(x)=0, (x+3)(x-1)=0$$

$$\therefore x = -3, 1$$

$$\therefore \text{Intercepts at } (0, -3), (-3, 0), (1, 0) \quad b) x = \frac{4 \pm \sqrt{16+96}}{6}$$

$$ii) \text{ vertex at } x = -\frac{b}{2a}$$

$$\therefore x = \frac{-2}{2}$$

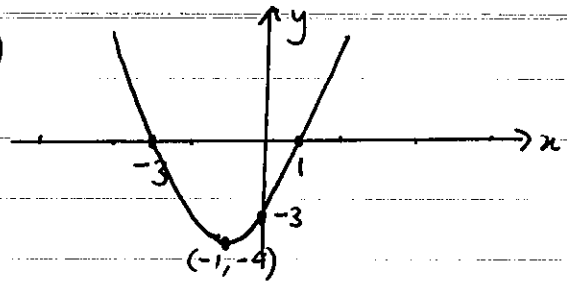
$$\therefore x = -1$$

$$\text{When } x=-1, f(x) = (-1)^2 + 2(-1) - 3$$

$$\therefore f(x) = -4$$

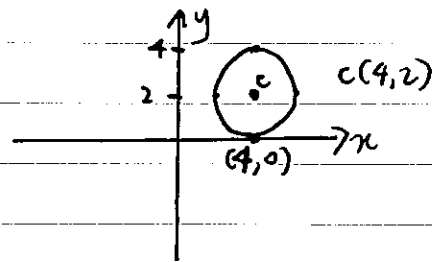
$$\therefore \text{minimum value is } -4$$

iii)



iv) curve decreasing when $x < -1$

e) Locus is a circle with $c(4, 2)$ and radius 2 units



$$f) f(x) = x^2 + 3x - 10$$

$$i) f'(x) = 2x + 3$$

$$ii) f'(x) = 0 \text{ when } 2x + 3 = 0$$

$$\therefore x = -\frac{3}{2}$$

Question 10

$$a) \tan 300^\circ$$

$$= -\tan 60^\circ$$

$$= -\sqrt{3}$$

$$b = -4$$

$$a = 3$$

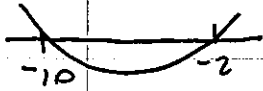
$$-4ac = 96$$

$$\therefore -12c = 96$$

$$\therefore c = -8$$

$$\therefore \text{eqn: } 3x^2 - 4x - 8 = 0$$

$$c) i) (x+10)(x+2) \geq 0$$



By inspection, $x \leq -10, x \geq -2$

$$ii) \sqrt{x^2 + 12x + 20}$$

need domain $x^2 + 12x + 20 \geq 0$

$$\therefore \{x : x \leq -10, x \geq -2\}$$

d) i) D has y co-ordinate $y = 7$

D lies on AD

$\therefore$ lies on $2x - y - 3 = 0$

$$\therefore 2x - 7 - 3 = 0$$

$$2x - 10 = 0$$

$$x = 5$$

$\therefore$ D is the point (5, 7)

$$ii) m_{BC} = \frac{7-3}{2-0}$$

$$= \frac{4}{2}$$

$$= 2$$

$$m_{AD} = \frac{7+3}{5-0}$$

$$= \frac{10}{5}$$

$$= 2$$

$$\therefore m_{BC} = m_{AD}$$

$$\therefore BC \parallel AD$$

$\therefore$ ABCD is a trapezium

$$iii) d_{BC} = \sqrt{(7-3)^2 + (2-0)^2}$$

$$= \sqrt{20}$$

$$= 2\sqrt{5}$$

$$\begin{aligned} d_{AD} &= \sqrt{(5-0)^2 + (7+3)^2} \\ &= \sqrt{25 + 100} \\ &= \sqrt{125} \\ &= 5\sqrt{5} \end{aligned}$$

$$iv) d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$= \left| \frac{2(0) - 1(3) - 3}{\sqrt{(2)^2 + (-1)^2}} \right|$$

$$= \left| \frac{-6}{\sqrt{5}} \right|$$

$$= \frac{6}{\sqrt{5}} \text{ units as required}$$

$$v) A = \frac{1}{2}(a+b)h$$

$$= \frac{1}{2}(2\sqrt{5} + 5\sqrt{5}) \frac{6}{\sqrt{5}}$$

$$= \frac{21\sqrt{5}}{\sqrt{5}}$$

$$= 21 \text{ units}^2$$

Question 11

$$a) \frac{m^2 - 4}{mn} \times \frac{2m}{2m - 4}$$

$$\frac{(m-2)(m+2)}{mn} \times \frac{2m}{2(m-2)}$$

$$= \frac{m+2}{n}$$

$$b) x^2 + 6x + 11$$

$$\Delta = (6)^2 - 4(1)(11)$$

$$= 36 - 44$$

$$= -12$$

$$\therefore \Delta < 0$$

$\therefore$ no real roots

$\therefore$ doesn't cross x-axis

$\therefore$ definite

$$a > 0$$

$\therefore$ positive definite

c) $C(2, -3)$ $A(5, -7)$

Let B be the other point

$\therefore C$ is the midpoint of AB

$$\therefore 2 = \frac{5+x_2}{2} \quad -3 = \frac{-7+y_2}{2}$$

$$4 = 5+x_2 \quad -6 = -7+y_2$$

$$\therefore x_2 = -1 \quad \therefore y_2 = 1$$

$\therefore B$ is the point $(-1, 1)$

d) $3x^2 + 5x - 1 = 0$

i) $\alpha + \beta = -\frac{5}{3}$

ii) $\alpha\beta = -\frac{1}{3}$

iii) $(\alpha+1)(\beta+1)$

$$= \alpha\beta + \alpha + \beta + 1$$

$$= -\frac{1}{3} + -\frac{5}{3} + 1$$

$$= -1$$

e) i) $\frac{d}{dx} x^{-3}$

$$= -3x^{-4}$$

$$= -\frac{3}{x^4}$$

ii) $\frac{d}{dx} (x-3)(2x+1)$

$$= \frac{d}{dx} 2x^2 - 5x - 3$$

$$= 4x - 5$$

iii) $\frac{d}{dx} \frac{5x^6 - 7x^2}{x}$

$$= \frac{d}{dx} (5x^5 - 7x)$$

$$= 25x^4 - 7$$

Question 12

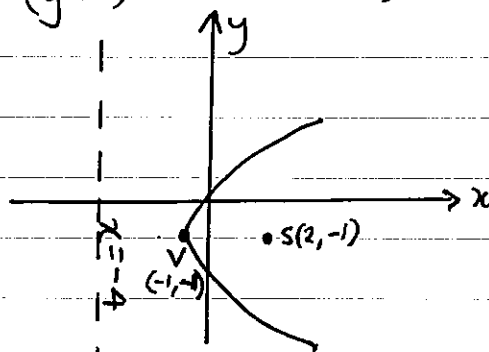
a) i) $S(2, -1)$ d: $x = -4$

concave right parabola

$V(-1, -1)$

ii) $(y+1)^2 = 4(3)(x+1)$

$$(y+1)^2 = 12(x+1)$$



b) $(x^2-2)^2 - 4(x^2-2) - 21 = 0$

let $u = x^2 - 2$

$$\therefore u^2 - 4u - 21 = 0$$

$$(u-7)(u+3) = 0$$

$$u = 7, -3$$

$$\therefore x^2 - 2 = 7 \quad \text{or} \quad x^2 - 2 = -3$$

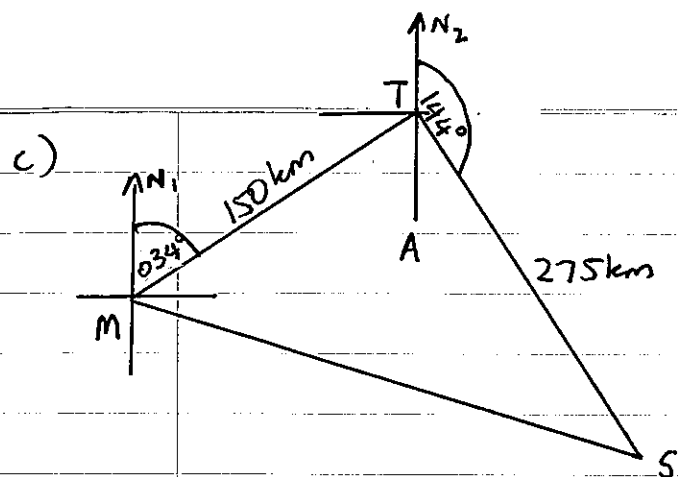
$$x^2 = 9$$

$$\therefore x = \pm 3$$

$$x^2 = -1$$

no real solutions

$$\therefore x = \pm 3$$



i) $\angle MTA = 34^\circ$ (alternate angles in parallel lines)

$\angle STA + 144 = 180$ (angle sum of straight angle)

$$\therefore \angle STA = 36$$

$$\therefore \angle MTS = 34 + 36$$

$$\therefore \angle MTS = 70^\circ \text{ as required}$$

$$\begin{aligned} \text{ii) } (MS)^2 &= (150)^2 + (275)^2 - 2(150)(275)\cos 70^\circ \\ &= 22500 + 75625 - 82500\cos 70^\circ \\ &= 98125 - 28216 \cdot 66182 \\ &= 69908 \cdot 33817 \end{aligned}$$

$$MS \approx 264.4 \text{ km}$$

$$\therefore MS = 264 \text{ km (to nearest km)}$$

$$\text{iii) } \frac{\sin \angle TMS}{275} = \frac{\sin 70^\circ}{MS}$$

$$\sin \angle TMS = 0.977358785$$

$$\angle TMS = 78^\circ \text{ (to nearest degree)}$$

$$\text{Bearing} = (034^\circ + 078^\circ)$$

$$\therefore \text{Bearing} = 112^\circ$$

d) $f(x) = x^2 - 1$

$$\begin{aligned} \text{i) } f(2) &= (2)^2 - 1 \\ &= 3 \end{aligned}$$

$$f(2+h) = (2+h)^2 - 1$$

$$\begin{aligned} f(2+h) &= 4 + 4h + h^2 - 1 \\ &= h^2 + 4h + 3 \end{aligned}$$

$$\text{ii) } \frac{f(2+h) - f(2)}{h}$$

$$= \frac{h^2 + 4h + 3 - 3}{h}$$

$$= \frac{h(h+4)}{h}$$

$$= 4 + h \text{ as required}$$

$$\text{iii) } f'(x) = 2x$$

$$\therefore f'(2) = 2(2) = 4$$

Question 13

$$\text{a) } 2x^2 - 3x - 4 \equiv A(x+2)^2 + B(x+2) + C$$

$$\begin{aligned} \text{When } x=0: 2(0)^2 - 3(0) - 4 &= A(0+2)^2 + B(0+2) + C \\ -4 &= 4A + 2B + C \end{aligned}$$

$$-4 = 4A + 2B + C$$

$$\text{When } x=-2: 2(-2)^2 - 3(-2) - 4 = A(-2+2)^2 + B(-2+2) + C$$

$$8 + 6 - 4 = C$$

$$\therefore \boxed{C = 10}$$

$$\therefore 4A + 2B + 10 = -4$$

$$4A + 2B = -14$$

$$\boxed{2A + B = -7} \Rightarrow \boxed{B = -7 - 2A}$$

$$\text{When } x=1: 2(1)^2 - 3(1) - 4 = A(1+2)^2 + B(1+2) + C$$

$$-5 = 9A + 3B + 10$$

$$\boxed{9A + 3B = -15}$$

$$\therefore \boxed{3A + B = -5} \Rightarrow \boxed{B = -5 - 3A}$$

$$\therefore -7 - 2A = -5 - 3A$$

$$A = 2$$

$$\therefore B = -11$$

$$\therefore A = 2, B = -11, C = 10$$

b) i) The function is not even as it doesn't have line symmetry about the y-axis

The function is not odd as it doesn't have point symmetry about the origin
 $\therefore$ The function is neither odd nor even

ii) $f(x)=0$ at a, c and e

iii) $f'(x)=0$ at b and d

iv) $f'(x)>0$ for $b < x < d$

Question 13

a) i) $\sin 30^\circ = \frac{4}{BD}$

$$\frac{1}{2} = \frac{4}{BD}$$

$$\therefore BD = 8 \text{ units}$$

ii) $\angle BDC = 60^\circ$ (angle sum $\triangle BDC$)

$\therefore \angle ADB + 60^\circ = 180^\circ$ (angle sum straight angle $\angle ADC$)

$$\therefore \angle ADB = 120^\circ$$

$$\frac{\sin x}{AB} = \frac{\sin 120^\circ}{13}$$

$$\sin x = \frac{AD \cdot \sqrt{3}}{2(13)}$$

$$\boxed{\sin x = \frac{\sqrt{3} AD}{26}}$$

$$(BC)^2 + (DC)^2 = (BD)^2 \text{ (by Pythagoras' Thm)}$$

$$(BC)^2 + 16 = 64$$

d) i) $y = x + m$ $x^2 + y^2 = 4$
 intersect when $x^2 + (x+m)^2 = 4$

$$x^2 + x^2 + 2mx + m^2 = 4$$

$$\therefore 2x^2 + 2mx + m^2 - 4 = 0$$

as required

ii) line will be a tangent with one point of intersection

$\therefore$ need $\Delta = 0$

$$\Delta = (2m)^2 - 4(2)(m^2 - 4)$$

$$= 4m^2 - 8m^2 + 32$$

$$= -4m^2 + 32$$

$$\text{for } \Delta = 0, -4m^2 + 32 = 0$$

$$\therefore 4m^2 = 32$$

$$m^2 = 8$$

$$m = \pm\sqrt{8}$$

$$\therefore m = \pm 2\sqrt{2}$$

$$\therefore (BC)^2 = 48$$

$$\therefore BC = \sqrt{48}$$

$$(BC)^2 + (AC)^2 = (AB)^2 \text{ (Pythag)}$$

$$\therefore (\sqrt{48})^2 + (AC)^2 = (13)^2$$

$$48 + (AC)^2 = 169$$

$$(AC)^2 = 121$$

$$\therefore AC = 11$$

$$\therefore AD = 11 - 4$$

$$\therefore AD = 7$$

$$\therefore \sin x = \frac{\sqrt{3} \cdot 7}{26}$$

$$\therefore \sin x = \frac{7\sqrt{3}}{26}$$